RESEARCH REPORT | REPORT 3

Hydraulic Fracturing and Unconventional Completion Design: Stimulated Volume, Depletion, and Interference

A technical review of fracture mechanics, proppant transport, cluster efficiency, decline behaviour, and parent-child interference

Scope: North American unconventional shale and tight reservoirs; practice as developed across the Barnett, Bakken, Marcellus, Eagle Ford, and Permian basins

Central conclusion

An unconventional well produces from the surface area it creates and connects, not from the volume of fluid and proppant pumped into it. Completion design should be evaluated on contacted and propped surface area per unit cost and on the depletion it induces in offset wells, rather than on stage count, cluster count, or total proppant loading treated as ends in themselves.

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Executive summary

Unconventional reservoirs have matrix permeability in the nanodarcy to microdarcy range, meaning hydrocarbons cannot travel commercially relevant distances to a wellbore on geological timescales. The economic proposition is not the reservoir's ability to flow but the engineer's ability to place conductive fracture surface close enough to every point in the rock that the resulting drainage distance becomes traversable within the well's producing life. The flow problem is solved by geometry, not by permeability.

This reframes every completion decision. Stage length, cluster spacing, and lateral length determine how much rock lies within reach of a fracture. Proppant type, size, and concentration determine whether that fracture still conducts once the pumps stop and closure stress acts. Fluid system determines how far proppant travels and how much of the created fracture is actually propped. The persistent industry difficulty is that created fracture volume, which diagnostics can estimate, and effective propped conductive area, which drives production, are not the same quantity and can diverge widely.

Cluster efficiency is the clearest expression of the gap. Production-log and fibre-optic diagnostic studies across multiple basins have repeatedly found that a substantial fraction of perforation clusters contribute little or no measurable flow, with commonly reported effective fractions falling well below 100% and in many documented cases below two-thirds. Fluid entering a stage distributes according to the path of least resistance, and stress shadowing from adjacent growing fractures actively concentrates it into the heel-side and toe-side clusters. Limited-entry design, using restricted perforation diameter and count to impose a deliberate friction drop, exists to force distribution against this tendency.

Design principle

Measure what the completion contacted, not what it consumed. Stage count, cluster count, and pounds of proppant are inputs; contacted and propped surface area, cluster contribution uniformity, and induced offset depletion are outcomes. A design optimised on inputs will reliably converge on pumping more of everything, which is the failure mode the diagnostic record most consistently documents.

Key findings

- Fracture orientation is set by the in-situ stress state, not by engineering preference: hydraulic fractures open against the minimum principal stress and propagate in the plane perpendicular to it. Horizontal laterals are drilled along minimum horizontal stress to generate transverse fractures.
- Cluster efficiency is routinely well below unity. Diagnostic studies across basins consistently report a substantial fraction of clusters contributing little or no flow, making limited-entry perforation design a first-order lever rather than a refinement.
- Proppant must survive closure stress at reservoir depth. Sand crushes progressively above roughly 6,000 psi closure, and crushed fines migrate and plug the very conductivity the proppant was placed to preserve.
- Slickwater and crosslinked gel trade proppant transport against fracture complexity and residual damage; the hybrid designs that dominate current practice are an explicit attempt to capture both.
- Unconventional wells exhibit Arps b-factors above 1 during transient linear flow, which mathematically implies infinite cumulative recovery if extrapolated without a terminal decline constraint.
- Parent-child interference is a depletion phenomenon, not a geometry accident: producing a parent well alters the local stress field, and child fractures grow asymmetrically toward the depleted low-stress region.

1. Stress state and fracture orientation

A hydraulic fracture opens along the path of least resistance, which means it opens against the smallest of the three principal stresses and propagates in the plane perpendicular to that direction. At depths typical of unconventional development, the vertical stress from overburden weight generally exceeds both horizontal stresses, so the minimum principal stress is horizontal and fractures are vertical, propagating along the azimuth of maximum horizontal stress. This is a consequence of the earth's stress state, and no completion design decision overrides it.

That constraint dictates well trajectory. To intersect the maximum number of transverse fractures, the lateral is drilled parallel to minimum horizontal stress, so that each fracture opens perpendicular to the wellbore. A lateral drilled along maximum horizontal stress instead generates longitudinal fractures running along the wellbore, which contact far less rock per unit of pumping. Determining stress azimuth, through image logs, borehole breakout analysis, or regional stress maps, is therefore a prerequisite to well planning rather than a post-drill observation.

Minimum horizontal stress magnitude is measured directly by diagnostic fracture injection test, in which a small volume is injected and the pressure decline analysed to identify fracture closure. The test also yields reservoir pressure and an estimate of permeability, making it the single most informative small-scale measurement available before full-scale stimulation. Stress contrast between the target interval and bounding layers is what determines height containment, and where that contrast is weak, height growth consumes energy in non-productive rock and can, in the worst case, connect the treatment to overlying water.

2. Fracture propagation models and their assumptions

Analytical fracture models remain useful for reasoning and for bounding expectations, provided their assumptions are stated. The PKN model assumes fracture height is fixed and length greatly exceeds height, producing an elliptical cross-section with maximum width at the centre; it is appropriate for well-contained fractures in a bounded interval. The KGD model assumes plane strain in the horizontal plane and applies where height exceeds length, which describes the early-time behaviour near the wellbore. The radial or penny-shaped model applies where the fracture grows unconstrained in all directions from a point source, as in a thick homogeneous interval with no stress barriers.

All three assume a single planar bi-wing fracture in a homogeneous linear-elastic medium. Microseismic monitoring in shale has demonstrated repeatedly that real stimulations frequently produce complex, branching networks interacting with pre-existing natural fractures and fabric. This is the origin of the stimulated reservoir volume concept, which describes the microseismically illuminated rock volume. SRV must be treated with care: microseismic events indicate shear failure somewhere in the rock, which is evidence that stress changed there, not evidence that a conductive propped pathway exists there. Illuminated volume systematically exceeds effective propped volume, and conflating them is a well-documented route to over-optimistic forecasts.

Net pressure behaviour during pumping carries interpretable diagnostic information. Nolte-Smith analysis plots log net pressure against log time, and the slope distinguishes confined extension from height growth, restricted extension, and tip screenout. A screenout, in which proppant bridges at the tip and arrests length growth while continued pumping inflates width, is a failure in a design pursuing length and a deliberate objective in a design pursuing near-wellbore conductivity in a soft high-permeability formation.

Table 1. Fracture propagation models and their validity domains

Model	Core assumption	Applies when	Known limitation
PKN	Fixed height; length >> height; plane strain in vertical plane	Well-contained fracture between competent stress barriers	Cannot represent height growth; invalid at early time near wellbore
KGD	Plane strain in horizontal plane; height >> length	Early-time near-wellbore growth; short fractures in thick intervals	Unrealistic for long fractures; over-predicts width at length
Radial / penny	Unconstrained growth from point source; isotropy	Thick homogeneous interval lacking stress barriers	Ignores layering entirely; rare in bedded shale sequences
Pseudo-3D	Layered stress profile with simplified height calculation	Routine commercial design across layered sequences	Height growth simplified; complexity and branching not represented
Planar 3D	Full 2D fluid flow within a planar fracture	Complex layering where height is a design-critical uncertainty	Computationally heavy; still assumes a single planar fracture
Discrete fracture network	Explicit interaction with natural fracture sets	Complex naturally fractured shale where branching dominates	Requires natural fracture characterisation that is rarely available

Model choice encodes an assumption about fracture geometry. A model that cannot represent the mechanism controlling a given treatment will produce a confident and precisely wrong answer, which is more dangerous than no answer.

3. Proppant selection and conductivity under stress

A fracture without proppant closes when pumping stops and contributes nothing. Proppant holds it open, and the governing property is fracture conductivity, the product of propped permeability and propped width. Conductivity is not a fixed catalogue value: it is the conductivity that survives closure stress, embedment into soft rock faces, cyclic stress from drawdown and shut-in, fines migration, and long-term diagenetic effects at reservoir temperature.

Closure stress at depth is the selection driver. Northern white sand performs adequately at moderate closure but crushes progressively above roughly 6,000 psi, and the crushed fines are worse than an absence of proppant because they migrate into the pack and plug the flow paths that remain. Resin-coated sand extends the range by encapsulating grains and containing fines when crushing does occur. Ceramic proppants, whether intermediate or high strength, are engineered for the highest closure environments and offer sphericity and crush resistance that sand cannot match, at a cost multiple that has to be justified against the incremental recovery it delivers rather than against its technical superiority in isolation.

Dimensionless fracture conductivity, which compares the fracture's capacity to carry fluid along its length against the formation's capacity to deliver fluid into it, is the parameter that determines whether conductivity is actually the binding constraint. Above roughly 10 to 30 depending on the analysis framework, additional conductivity yields negligible incremental production because the reservoir cannot supply fluid fast enough to use it, and further investment in premium proppant is spending on a constraint that is not binding. In nanodarcy shale this threshold is reached at modest conductivity, which is the technical reason sand dominates shale completions despite ceramic's superior specifications.

Table 2. Proppant classes and application envelopes

Proppant class	Closure stress envelope	Relative cost	Governing consideration
Northern white sand	Moderate; degrades progressively above ~6,000 psi	Lowest	Crush generates migrating fines that plug the remaining pack
Regional / in-basin sand	Lower than northern white at equal mesh	Lowest delivered	Logistics economics often outweigh the conductivity penalty
Resin-coated sand	Extends sand range; contains fines on crush	Intermediate	Resin curing is temperature-dependent; can interact with fluid chemistry
Intermediate-strength ceramic	High closure stress service	High	Justified only where conductivity is demonstrably the binding constraint
High-strength ceramic (bauxite)	Highest closure; deep, high-stress reservoirs	Highest	Rarely justified in shale where reservoir deliverability binds first
Lightweight ceramic	Moderate to high; density near sand	High	Improves transport in low-viscosity slickwater; cost limits routine use

Stress envelopes are indicative and depend on mesh size, grain quality, and test standard. API RP 19C and RP 19D define the measurement conditions; conductivity quoted without its test standard, stress, and temperature is not comparable across suppliers.

4. Fluid systems and the transport-versus-complexity trade

The fracturing fluid must transport proppant into the fracture and then get out of the way. These requirements conflict, and the fluid systems in use represent different resolutions of the conflict rather than a progression from worse to better.

Slickwater is water with a friction-reducing polymer at low concentration. Its viscosity is near water's, so it carries proppant poorly and relies on high pump rate and turbulence to keep grains suspended, producing settling and a proppant bank along the fracture bottom. In exchange, its low viscosity promotes complex branching fracture networks in naturally fractured shale, it leaves minimal residue, and it is inexpensive. Crosslinked gel inverts every term: excellent transport and near-uniform proppant distribution, at the cost of planar rather than complex geometry, expensive chemistry, and polymer residue that damages the proppant pack unless the breaker schedule performs correctly at reservoir temperature.

Hybrid designs attempt to capture both: slickwater early to initiate and extend complexity, then progressively viscosified stages to place proppant in the created network. The residual difficulty is that the complexity created by the low-viscosity stage is precisely the geometry the viscous stage struggles to reach, because narrow branching apertures screen out. Created complexity and propped complexity remain distinct quantities, and no fluid programme has fully closed that gap.

5. Cluster efficiency and limited entry

Within a stage, fluid distributes among perforation clusters according to the resistance each presents. Absent deliberate intervention, distribution is markedly non-uniform. Production logging and fibre-optic distributed acoustic sensing studies across multiple basins have found repeatedly that a substantial fraction of clusters contribute little or no measurable flow. The consequence is direct: a stage nominally containing several clusters may be producing from a minority of them, and the rock the silent clusters were placed to contact is simply not drained.

Two mechanisms drive this. Stress shadowing means the first fracture to open raises local stress and makes it mechanically harder for its neighbours to open, so fluid preferentially feeds the outermost clusters where the shadow is weakest. Perforation friction and near-wellbore tortuosity vary between clusters, and small differences are amplified because flow rate depends nonlinearly on the resistance. Limited-entry design counters both by restricting perforation diameter and count so that perforation friction dominates the total pressure drop; when that friction is large relative to the differences between clusters, distribution is forced toward uniformity. Extreme limited entry pushes this further, accepting substantial treating pressure and perforation erosion in exchange for distribution control.

6. Decline behaviour and the b-factor problem

Unconventional wells decline in a way conventional decline analysis was not built to describe. Early production is dominated by transient linear flow, in which the pressure disturbance propagates perpendicular to the fracture face and has not yet encountered any boundary. During this regime, rate declines approximately with the inverse square root of time, and the decline is extremely steep: substantial fractions of first-year production are lost by year two in typical shale wells.

Fitting Arps hyperbolic decline to that regime produces b-factors above 1, which conventional reservoirs essentially never exhibit. This is a mathematical hazard, not merely a curiosity. The Arps hyperbolic form with b greater than or equal to 1 integrates to infinite cumulative production over infinite time, so a b-factor above 1 extrapolated without constraint forecasts unbounded reserves. The standard remedy is a modified hyperbolic with an imposed terminal exponential decline, switching to a fixed minimum decline rate once the hyperbolic falls below it. The terminal rate is an assumption, frequently a consequential one, and it should be stated as such rather than buried in a spreadsheet default.

Physically motivated alternatives exist because the Arps form is empirical and was derived for boundary-dominated flow in conventional reservoirs. Duong's method is constructed specifically for fracture-dominated linear flow. Rate-transient analysis using linear-flow diagnostics can extract effective fracture half-length and contacted area, which is more useful than a curve fit because those parameters connect the decline to the completion decisions that produced it and therefore transfer to the next well.

Table 3. Flow regimes in a fractured unconventional well

Flow regime	Physical description	Diagnostic signature	Engineering implication
Fracture linear	Flow within the fracture toward the wellbore	Very early; often masked by cleanup and flowback	Rarely interpretable in practice; short-lived
Bilinear	Simultaneous flow in fracture and matrix	Quarter-slope on log-log rate or pressure derivative	Indicates finite fracture conductivity is limiting
Transient linear (matrix)	Matrix flow perpendicular to fracture face	Half-slope; dominant regime in shale; can persist for years	Contacted area, not permeability, controls rate; source of b > 1
Compound / transitional	Interference between adjacent fractures	Departure from half-slope behaviour	Marks the end of independent fracture drainage; spacing is now binding

Flow regime	Physical description	Diagnostic signature	Engineering implication
Boundary-dominated	All boundaries felt; depletion of a closed volume	Unit slope on the derivative; exponential-like decline	Only here does conventional EUR logic apply; often decades away

Many shale wells never reach boundary-dominated flow within their economic life. Applying decline logic that presupposes that regime to a well still in transient linear flow is a category error, not a conservative approximation.

7. Well spacing and parent-child interference

Well spacing in unconventional development is the central economic decision, and it is genuinely two-sided. Space too widely and rock between the wells is never drained, leaving recoverable oil stranded because no fracture reaches it. Space too tightly and wells share the same drainage volume, so total recovery per section rises little while capital per barrel rises steeply. The optimum depends on effective fracture half-length, which is not directly observable and must be inferred from rate-transient analysis, interference tests, or tracer studies.

Parent-child interference has proven the harder problem, and it is fundamentally geomechanical rather than geometric. A parent well producing for months or years depletes pressure in the rock around it, and because pore pressure supports part of the total stress, depletion reduces the local minimum horizontal stress. When a child well is subsequently fractured nearby, its fractures encounter that depleted low-stress region and grow preferentially toward it, since fractures follow the path of least resistance. The child's stimulation is therefore asymmetric and biased into rock the parent has already drained, and in the adverse case the child's fractures connect directly to the parent, creating a frac hit that can damage the parent's productivity while delivering an underperforming child.

The timing dependence follows directly from the mechanism. The stress perturbation grows as depletion accumulates, so interference severity increases with the delay between parent and child. This is the technical basis for co-development, completing all wells in a section together before meaningful depletion establishes a stress gradient. Mitigations for infill into depleted rock, including pre-loading the parent with water to re-pressurise the near-well region and modifying the child's design near the parent, are partial. They manage a problem that co-development avoids, and their inconsistent field results across basins reflect that distinction.

8. Recommendations

- Determine stress azimuth and magnitude before well planning, using image logs, breakout analysis, and diagnostic fracture injection tests, and treat the results as trajectory constraints rather than post-drill observations.
- Evaluate completion designs on contacted and propped surface area per unit cost and on measured cluster contribution, not on stage count, cluster count, or proppant loading as standalone metrics.
- Instrument a representative subset of wells with fibre-optic or production-log diagnostics and design limited-entry perforation schemes against the measured cluster efficiency rather than a basin default.
- Select proppant against the dimensionless fracture conductivity at which the reservoir stops being able to supply fluid, and justify premium proppant only where conductivity is demonstrably the binding constraint.
- Constrain every hyperbolic decline fit with an explicit and stated terminal decline rate; never report an EUR from an unconstrained fit with a b-factor at or above 1.
- Prefer rate-transient analysis over curve fitting where data permit, because effective half-length and contacted area transfer to future completion decisions in a way that a b-factor does not.
- Co-develop sections where capital allocation permits, and treat infill into materially depleted rock as a geomechanically different problem requiring its own design rather than a repeat of the parent's.

- Report SRV and propped effective volume as separate quantities in every forecast, with the diagnostic basis for each stated.

9. Limitations

This report describes general practice developed across North American unconventional basins and is not prescriptive for any specific asset. Stress states, natural fracture systems, thermal maturity, and fluid phase behaviour vary substantially between and within basins, and completion designs that perform in one do not transfer reliably to another. Cluster efficiency figures reflect diagnostic studies of instrumented wells, which are not a random sample of the well population. Decline behaviour is reported qualitatively by flow regime because published shale decline parameters vary widely with area, vintage, completion design, and operator practice, and quoting a single representative value would misrepresent that spread.

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